

Zad.1 Oblicz $x + |x|$ oraz $x - 2|x|$.

a) $x = -3$ b) $x = 6\sqrt{2} - 8$ c) $x = \pi - 2\sqrt{3}$

1) Określamy znak z jakim opuszczamy wart. bezwzgl. dla danego x i obliczamy wyrażenia:

a) $x = -3$

$$x < 0, \text{ więc } |x| = -x$$

$$|x| = |-3| = -(-3) = 3$$

$$x + |x| = x + (-x) = 0$$

$$x - 2|x| = x - 2(-x) = 3x = -9$$

Odp. a) $x + |x| = 0$, $x - 2|x| = -9$.

b) $x = 6\sqrt{2} - 8$

$$6\sqrt{2} - 8 \approx 0,49 > 0$$

$$x > 0, \text{ więc } |x| = x$$

$$|x| = 6\sqrt{2} - 8$$

$$x + |x| = x + x = 2x = 12\sqrt{2} - 16$$

$$x - 2|x| = x - 2x = -x = -(6\sqrt{2} - 8) = -6\sqrt{2} + 8$$

Odp. b) $x + |x| = 12\sqrt{2} - 16$, $x - 2|x| = -6\sqrt{2} + 8$.

c) $x = \pi - 2\sqrt{3}$

$$\pi - 2\sqrt{3} \approx -0,32 < 0$$

$$x < 0, \text{ więc } |x| = -x$$

$$|x| = -(\pi - 2\sqrt{3}) = -\pi + 2\sqrt{3}$$

$$x + |x| = x + (-x) = 0$$

$$x - 2|x| = x - 2(-x) = 3x = 3\pi - 6\sqrt{3}$$

Odp. c) $x + |x| = 0$, $x - 2|x| = 3\pi - 6\sqrt{3}$.